

DaSEH Day 9 Cheatsheet

Functions

Major concepts

- **Create functions to:**
 - Do custom actions
 - Reduce repetitive code
 - Organize code into manageable chunks
 - Avoid running code unintentionally
 - Use names that make sense to you
- **Custom function syntax:**

```
function_name <- function(inputs) {  
  # Document inputs here  
  <function body>  
  return(value)  
}
```

Examples

Example of an **anonymous function** used with `across()`:

```
iris |>  
  summarize(across(  
    starts_with("Sepal"),  
    \(x) mean(x, na.rm = TRUE)  
  ))
```

Functions

Library/package	Piece of code	Example of usage	What it does
Base R	<code>function()</code> or <code>\()</code>	<code>div_100 <- function(x) x / 100</code> or <code>div_100 <- \ (x) x / 100</code>	Creates a function; here, x is divided by 100.
Base R	<code>lapply()</code>	<code>lapply(some_list, a_function)</code>	Applies a function over a list or vector.
Base R	<code>sapply()</code>	<code>sapply(some_list, a_function)</code>	Applies a function over a list or vector; returns a matrix or vector by default.
tidyverse (dplyr)	<code>across()</code>	<code>iris ></code> <code> mutate(across(</code> <code> contains("Sepal"), round</code> <code>))</code> or <code>iris ></code> <code> group_by(Species) ></code> <code> summarise(across(</code> <code> starts_with("Sepal"),</code> <code> \ (x) mean(x, na.rm = TRUE)</code> <code>))</code>	Applies a function across columns in a data frame.
tidyverse (dplyr)	<code>mutate_if()</code>	<code>iris > mutate_if(is.numeric, round)</code>	Applies a function to selected columns in a data frame.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.